

2c. Content of Physiological factors affecting performance (H555/01)

Component 01, Physiological factors affecting performance, focuses on developing the learner's knowledge of the science behind physical activity. This includes the structure and function of key systems in the human body, the forces that act upon us and the adaptations we make to our bodies through diet and training regimes.

Through the study of this component, learners will gain a deeper understanding of key systems in the body and how they react to changes in diet and exercise. They will also study the effects of force and motion on the body and how these effects can be used in physical activities to our advantage.

This topic focuses on key systems of the human body involved in movement and physical activity.

Learners will develop their knowledge and understanding of the changes within these body systems prior to exercise, during exercise of differing intensities and during recovery.

Learners will know and understand the different energy systems and factors that affect the interplay of the energy systems during physical activity.

Application of this theoretical knowledge will enable learners to understand how changes in physiological states can influence performance in physical activities and sport.

1.1.a. Skeletal and muscular systems

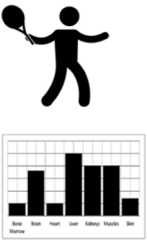
Learners will develop their knowledge and understanding of the roles of the skeletal and muscular systems in the performance of movement skills in physical activities and sport.

Knowledge and understanding of the skeletal system is required and should include the structure and functions of bones, joints and connective tissues.

Knowledge and understanding of planes of movement, the roles of muscles and types of contraction will be developed. Learners will also be

able to analyse movement in physical activities and sport applying the underlying knowledge of muscular contraction.

Topic Area	Content
Joints, movements and muscles 	• shoulder: ◦ flexion, extension, abduction, adduction, horizontal flexion/extension, medial and lateral rotation, circumduction ◦ deltoid, latissimus dorsi, pectoralis major, trapezius, teres minor • elbow: ◦ flexion, extension ◦ biceps brachii, triceps brachii • wrist: ◦ flexion, extension ◦ wrist flexors, wrist extensors • hip: ◦ flexion, extension, abduction, adduction, medial and lateral rotation ◦ iliopsoas, gluteus maximus, medius and minimus, adductor longus, brevis and magnus • knee: ◦ flexion, extension ◦ hamstring group: biceps femoris, semi-membranosus, semi-tendinosus ◦ quadriceps group: rectus femoris, vastus lateralis, vastus intermedius and vastus medialis • ankle: ◦ dorsi flexion, plantar flexion ◦ tibialis anterior, soleus, gastrocnemius • planes of movement: ◦ frontal ◦ transverse ◦ sagittal.
Functional roles of muscles and types of contraction 	• roles of muscles: ◦ agonist ◦ antagonist ◦ fixator • types of contraction: ◦ isotonic ◦ concentric ◦ eccentric ◦ isometric.

Topic Area	Content
Analysis of movement  The bar chart shows energy expenditure for various activities: Rest (low), Sleep (low), Heart (medium), Jog (high), Walking (medium), and Sit (low). The 'Jog' activity shows the highest energy expenditure.	- analyse movement with reference to: - joint type - movement produced - agonist and antagonist muscles involved - type of muscle contraction taking place.
Skeletal muscle contraction 	- structure and role of motor units in skeletal muscle contraction - nervous stimulation of the motor unit: - motor neuron - action potential - neurotransmitter 'all or none' law.
Muscle contraction during exercise of differing intensities and during recovery 	- muscle fibre types: - slow oxidative - fast oxidative glycolytic - fast glycolytic - recruitment of different fibre types during exercise of differing intensities and during recovery.